

Employees Classification Project | Salifort Motors

Executive Summary

ISSUE / PROBLEM

Leadership team is concerned about how many employees are leaving the company. There is a high rate of turnover which is costly in the financial sense. Salifort makes a big investment in recruiting, training, and upskilling its employees.

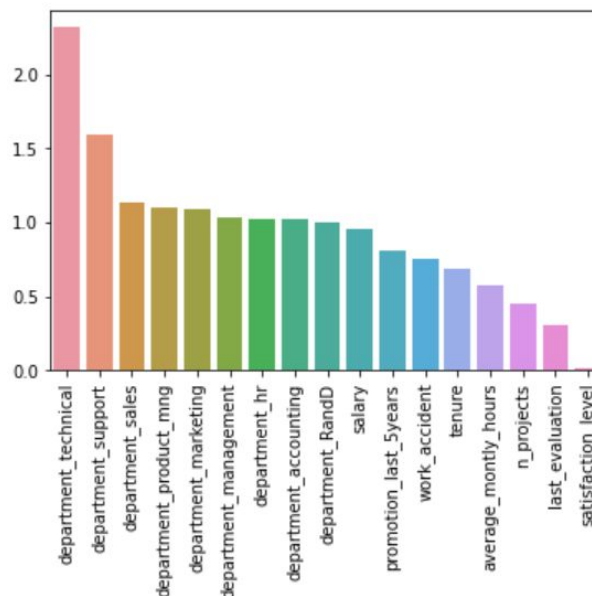
RESPONSE

Predicting whether an employee will leave the company, and discovering reasons behind their departure, we could understand the problem and develop a solution. Human Resources surveyed a sample of employees to learn more about what might be driving turnover. Then, I was tasked to design a model that predicts whether an employee will leave the company based on their data records. Hence, a logistic regression model was developed.

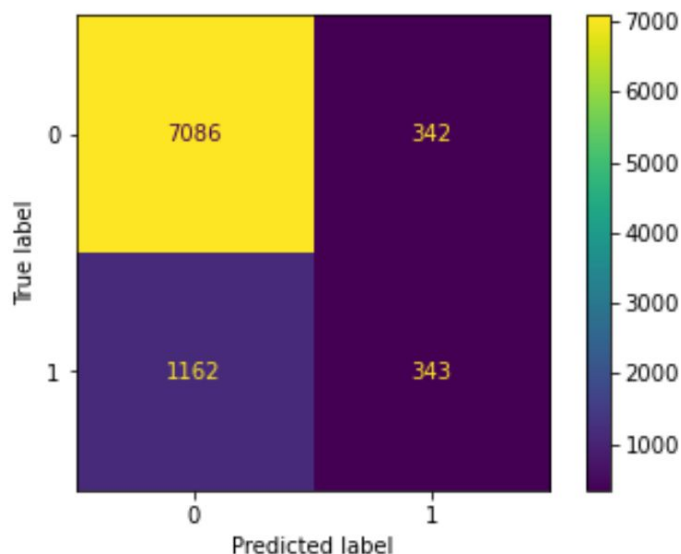
IMPACT

The model could help the company increase retention and job satisfaction for current employees, and save money and time training new employees. However, we should keep in mind the possibility of predicting false positive and false negative classifications.

Figure shows the importance of different features on model's prediction



Confusion matrix shows the model high tendency of making false negative errors (where Left= 1, Retained = 0)



KEY INSIGHTS

- We recommend that the company modifies its policies in a way that minimize the effects mentioned in the above figure. And to seek for increasing fairness between different departments since some departments tend to have higher turnover rate than others.
- The model achieved 83% accuracy, 80% precision, 83% recall, and 80% f1-score, even though the results would be much lower if we give more weight for the goal of predicting employees that WILL leave the company.
- next step could be to develop an XGBoost or Random Forest model to increase the reliness of the predictions.